Shivshankar Chaudhary

Full stack Python AI Engineer

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SUMMARY

AI and Digital Transformation Consultant with 8 years of AI and deep expertise in Generative AI, Deep Learning, and AI Agents, delivering strategic and scalable solutions across manufacturing, financial services, automotive, and health insurance sectors & Banking domain safety mechanisms.

EXPERIENCE

AI Engineer (Agentic AI & Generative AI specialization)

TCS 02/2026 - Present Bangalore, India

- Developed a BRS-ERP Placement Module and PRD Business Requirements Chatbot using Python, LangChain, LangGraph, PostgreSQL, and LLMs.
- Built conversational AI solutions integrating OpenAI ChatGPT and Anthropic APIs, with Ollama for local/private LLM deployment and experimentation.
- Designed LLM-based business requirement and document-processing workflows to automate requirement analysis, extraction, and response generation.
- Developed agentic AI automation workflows using n8n, integrating LLM agents, APIs, business processes, and external systems.
- Built an AI-powered e-commerce solution incorporating Computer Vision for intelligent image-based product processing and analysis.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using LangChain, embeddings, vector databases (FAISS and Pinecone), and LLMs to provide context-aware responses.
- Applied LSTM-based deep learning models for sequence-based prediction and analysis use cases.
- Built batch and real-time data ingestion pipelines using Apache Spark and Apache Kafka for scalable data processing and streaming.
- Containerized applications using Docker and deployed/scaled services using Kubernetes.
- Used Git for source-code management, version control, collaboration, and CI/CD-oriented development workflows.
- Integrated GenAI, Agentic AI, data engineering, RAG, computer vision, and cloud-native technologies to develop scalable enterprise AI solutions.

Packaged app development lead (Ai) [GE Healthcare]

Accenture 12/2024 - 12/2025 Indore, India

- Image and video Recognition Convolution Neural network are extensively used in applications such as facial recognition, Autonomous driving and medical image analysis. NLP has Recurrent Neural Network and transformers power language Chatbot translation integrates machine learning and AI into virtual assistants, allowing them to instantly translate user messages and bot responses in real-time, and sentiment analysis. Finance predicting stock prices, fraud detection and risk management
- Own the end to end ML lifecycle, including data ingestion, feature engineering, training, evaluation, deployment, monitoring, retraining, and rollback
- Developed and deployed machine learning and deep learning solutions for image recognition, NLP, finance, and AI automation use cases.
- Built CNN-based computer vision models for applications including facial recognition, autonomous driving, and medical image analysis using Keras/TensorFlow.
- Implemented RNNs, GRUs, ANNs, and transformer-based architectures for NLP applications such as chatbots, language translation, and sentiment analysis.
- Developed real-time NLP translation pipelines to translate user messages and chatbot responses using machine learning and AI techniques.
- Applied machine learning to financial use cases, including stock-price prediction, fraud detection, and risk management. .MSI, .EXE Extenstion installer development

SKILLS

Programming & Data:

Python **Pandas** **Numpy** **Pytorch**
Tensorflow **Langchain** **Langgraph**
Vectordb **Rag** **K-NN** **SVM** **Yolo**
Docker/Kubernetes **FastAPI's** **Django**
ReactJs **SQL** **spaCy** **Keras**
N8N AI Automation **Scikit-learn** **Seaborn**
Matplotlib **GRUs** **ANNs**
GPU-Accelerated ML **AWS** **Azure** **GCP**
Git/GitHub **NLU** **Safty Guardrails** **NLP**
MCP **FNNs** **CNN** **RNNs** **GANs**
ML Pipelines **Airflow** **MLflow**
Feature Stores **MLOps** **Model Deployment**
Monitoring **Auto-Retraining** **Unreal-Engine 5**

Data Structures and Algorithms:

Python Fundamentals **Complexity Analysis**
Basic Data Structures/Arrays/List **Strings**
Hashing **Linked Lists** **Stack** **Queue**
Searching **Sorting** **Recursion G Backtracking**
Trees **Heap** **Graphs** **Greedy Algorithms**
Dynamic Programming **Bit Manipulation**
Advanced Algorithms **Advanced Data Structures**
Problem-Solving Patterns

System Design:

Scalability **Load Balancer** **Caching**
Database **SQL vs NoSQL** **Replication**
Sharding **Message Queue** **API Design**
Rate Limiting **Microservices** **CAP Theorem**

Object-Oriented Design / LLD/Computer Networks

SOLID principles **TCP vs UDP** **DNS**

EXPERIENCE

Python Developer

MDP Infra Pvt. Ltd. (Client: MPMKVVCL) 03/2023 - 08/2024 Bhopal M.P

- Technology solutions for utility management Developed secure reverse proxy hosting and multi- factor authentication (MFA) API with OCR computer vision AI visualization digital meter text reading qcportal .mpcz have domain Sampling process according to TKC vender, technology used python,django,and Fastapi and Postgresql based systems with frontend with ReactJS, application developed Chatbot using Langchain,and Langgraph with hosting Azure Function and Azure Cosmos DB is Microsoft's fully managed, globally distributed NoSQL database service
- Developed technology solutions for utility management, supporting secure, scalable, and data-driven operational workflows.
- Built and deployed a secure reverse proxy and Multi-Factor Authentication (MFA) API to strengthen application security and access control.
- Developed an OCR-based Computer Vision solution for automated digital meter text recognition and reading, improving meter-data processing efficiency.
- Designed and implemented sampling and quality-control workflows based on TKC vendor processes for utility operations.

Software Engineer [Client:(ESPN(Live broadcasting company)]

TeleDNA Communications Pvt Ltd 07/2013 - 03/2019 Bangalore, India

- Developed web scraping and automated data extraction pipelines using Python, BeautifulSoup, Pandas, NumPy, and Django.
- Implemented Multi-Factor Authentication (MFA) using face recognition, QR codes, OTP, and speech recognition to enhance application security.
- Developed and integrated a DNN-based face detection model using deep learning for real-time facial recognition and authentication.
- Integrated computer vision and AI-based authentication into web applications to support secure and automated user verification.
- Video and audio Packets are sent quickly.
- If a small packet is lost, the streaming may continue.
- Waiting to resend every lost Packet could cause delays.
- Low latency matters more than perfect delivery.
- Banking secter-TCP for secure, reliable application communication.
- E-comerce companies-TCP for web pages,orders,and payments.
- Gaming companies-UDP for real-time player movement and gameplay.
- Video conferencing companies- UDP often for real-time audio and video,with other protocols or TCP as fallback.

KEY ACHIEVEMENTS

Technology Development

- Developed and secured **enterprise ERP web applications** using **MCP-based architecture** and scalable **reverse proxy solutions**.
- Implemented **security guardrails, authentication, authorization, and access-control mechanisms** to protect enterprise applications and AI-enabled workflows.
- Integrated **MCP architecture and AI safety guardrails** to enable controlled, secure, and reliable interactions between AI systems, tools, and enterprise services.
- Optimized application architecture and infrastructure for **high performance, scalability, reliability, and secure production deployment**.
- Applied secure engineering practices across the **end-to-end ERP application lifecycle**, from development and integration through deployment and monitoring.
- Built a **Speech-to-Text application** using NVIDIA **Parakeet TDT 0.6B v2 ASR**.
- Developed an interactive **Streamlit UI** for audio upload, preview, and transcription.
- Implemented **GPU/CUDA detection** and **bfloat16 mixed precision** for faster inference.
- Designed **audio preprocessing pipeline** with resampling, mono conversion, and temporary-file handling.
- Implemented **chunking and optimized attention** for long audio files (>8 minutes).
- Generated **segment-level timestamps** and structured transcription results.
- Added **CSV, TXT, and SRT export** functionality.
- Implemented speech analytics including **word count, speech duration, and WPM**.
- Improved application reliability through **error handling, GPU memory cleanup, session history, and progress tracking**.

EDUCATION

M.tech

Rajiv Gandhi Proudyogiki Vishwavidyalaya

08/2019 - 02/2022 Indore

Bachelor of Engineering

Rajiv Gandhi Proudyogiki Vishwavidyalaya

04/2009 - 06/2013 Indore, India

Udemy Certificate Ai/ML

udemy.com/certificate/UC-045fe37e-1aba-4ed6 9b4a-ebf84afe1953

Research in M.Tech

A robust load frequency control in two-area power system with help of SCA algorithm Matlab software